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Data Science Fundamentals

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Before we get started…

Practicalities

Who am I?

- **Rik Tonnelier**
- **C# .NET developer** with an interest in data
- **Ex IMS** (former dutch counterpart of DSPS) student
- **Email**: rik.tonnelier@thomasmore.be



Who are you?

Let’s do a short round!

Say your **name** + **one boring fact** (e.g. I once wore funny socks as a joke. Now every year I get gifted funny socks and at this point I’m commited)



Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
  - https://mlu-explain.github.io/logistic-regression/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory exercises, sometimes in class)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression
Week - 6	Classification - kNN	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Lifesnaps guidance		

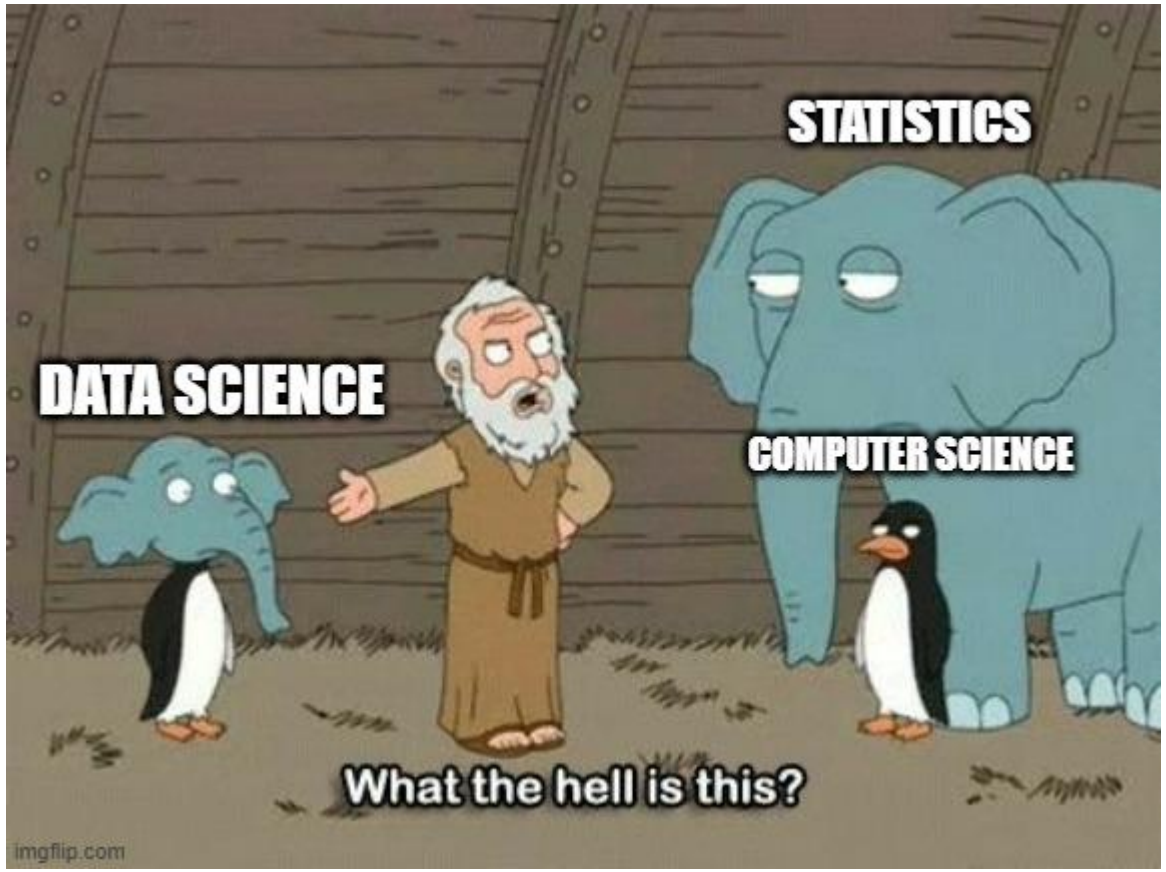
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Now we get to the fun part

Intro to Data Science

What is data science



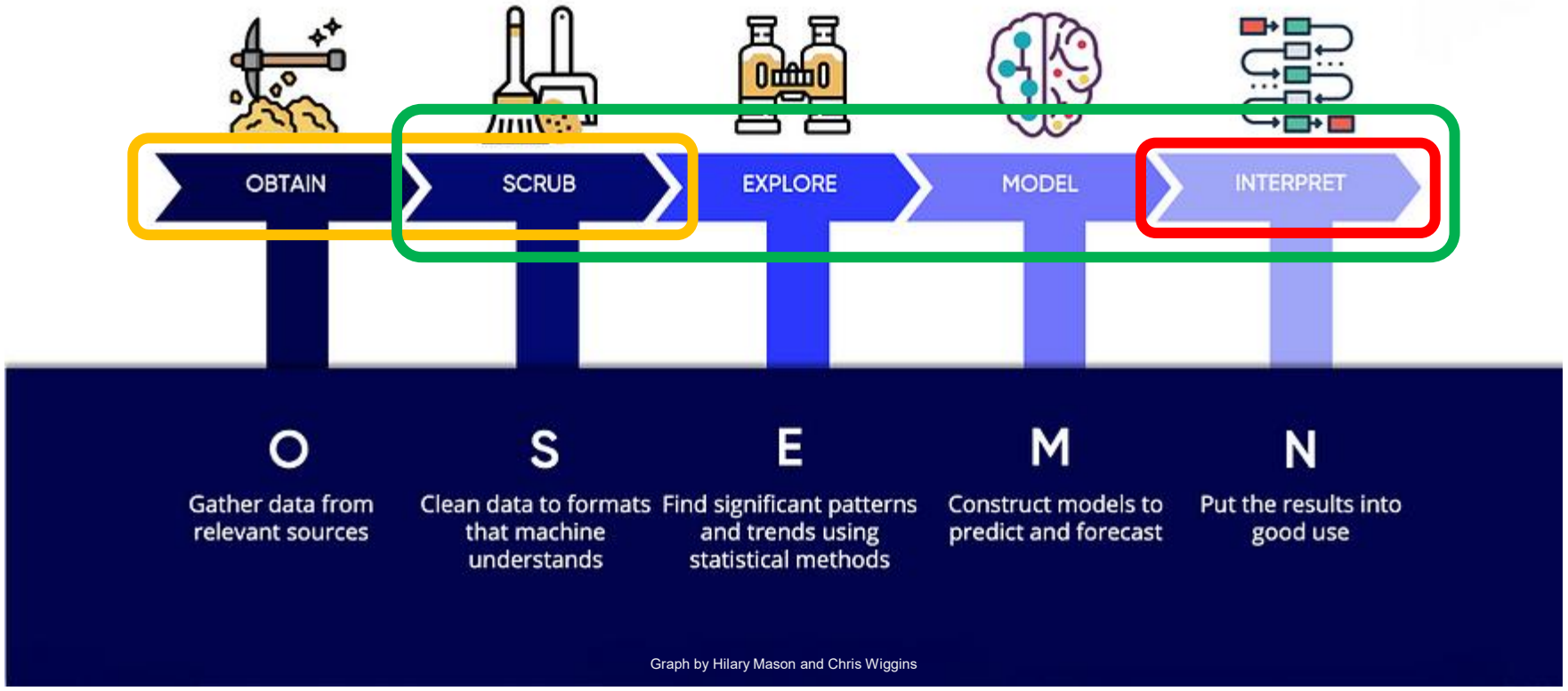
What is data science

- Data Science is about data gathering, analysis and decision-making.
- Data Science is about finding patterns in data, through analysis, and make future predictions.
- Data science combines from computer science and statistics to provide meaningful insights from large amounts of complex data.
- By using Data Science, companies are able to make:
  - Better decisions (should we choose A or B)
  - Predictive analysis (what will happen next?)
  - Pattern discoveries (find pattern, or maybe hidden information in the data)

Why is data science important

- Some examples:
  - To create promotional offers
  - To find the best suited time to deliver goods
  - To forecast the next years revenue for a company
  - To analyze health benefit of training
  - To predict event sign-ups

The OSEMN process



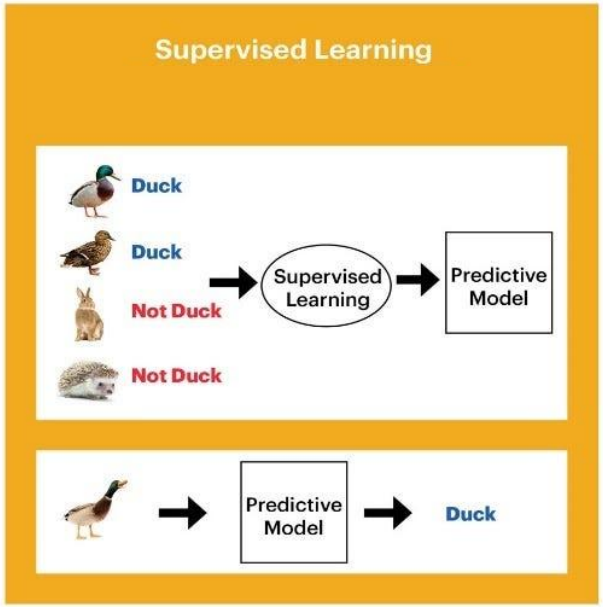
Legend:
Data engineer
Data analist
Data scientist

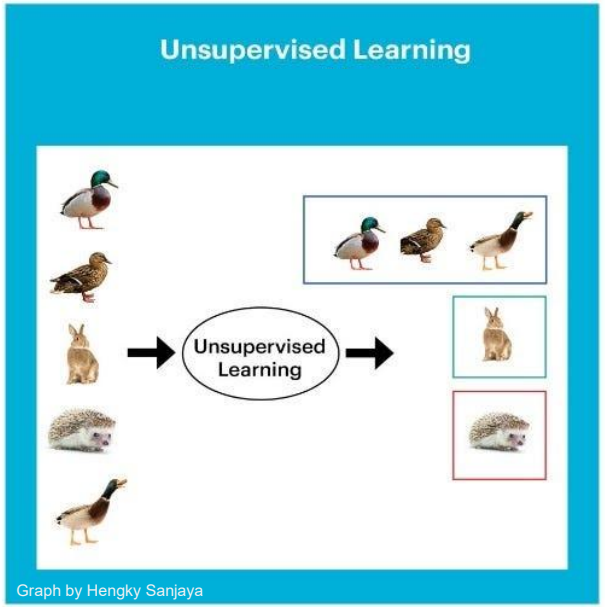
Machine learning

- An approach to achieve artificial intelligence through systems that can learn from experience to find patterns in a set of data.
- It relies on teaching a computer to recognize patterns by example, rather than programming it with specific rules
- A way to make predictions
  - 1. Takes in data
  - 2. Learns patterns from said data
  - 3. Classifies new data it has not seen before

Types of machine learning

- Supervised
  - A ML task where the program is provided with training data that is labeled.
  - Given the inputs, the system knows what the expected output label is.
- Unsupervised
  - A ML task where the program is provided with training data that is unlabeled.
  - Given the inputs, we do not know what the output should be.





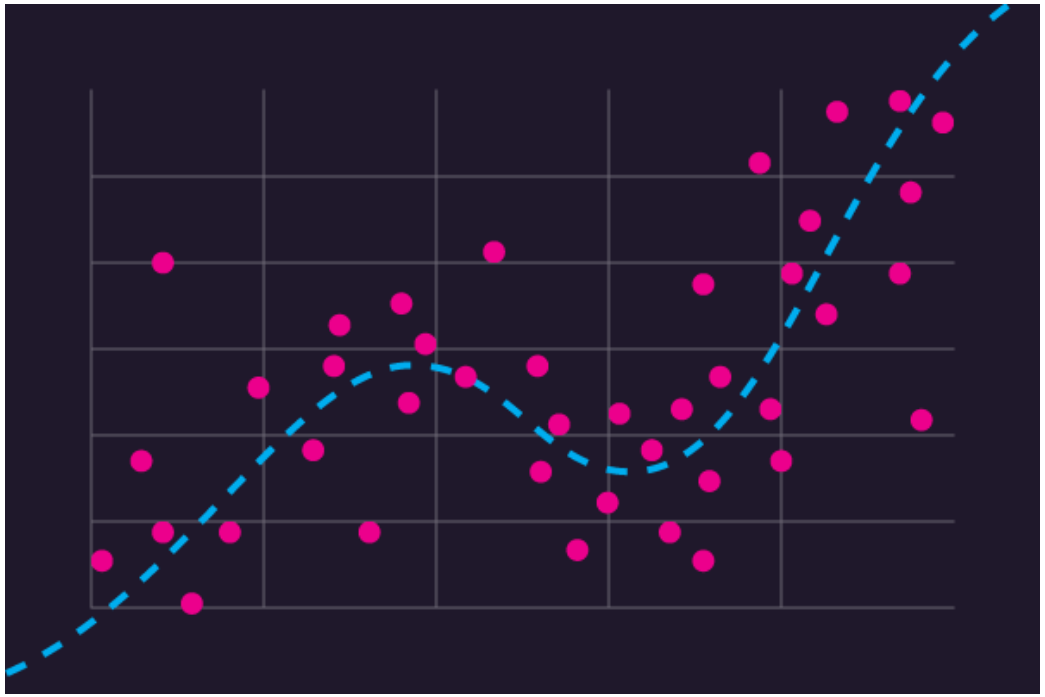
Western Digital.

Regression

- A method of machine learning wherein the variable we wish to predict (the dependent variable) is of a continuous nature.
- The value of a given entry is determined based on known cases.

Supervised or unsupervised?

Supervised!

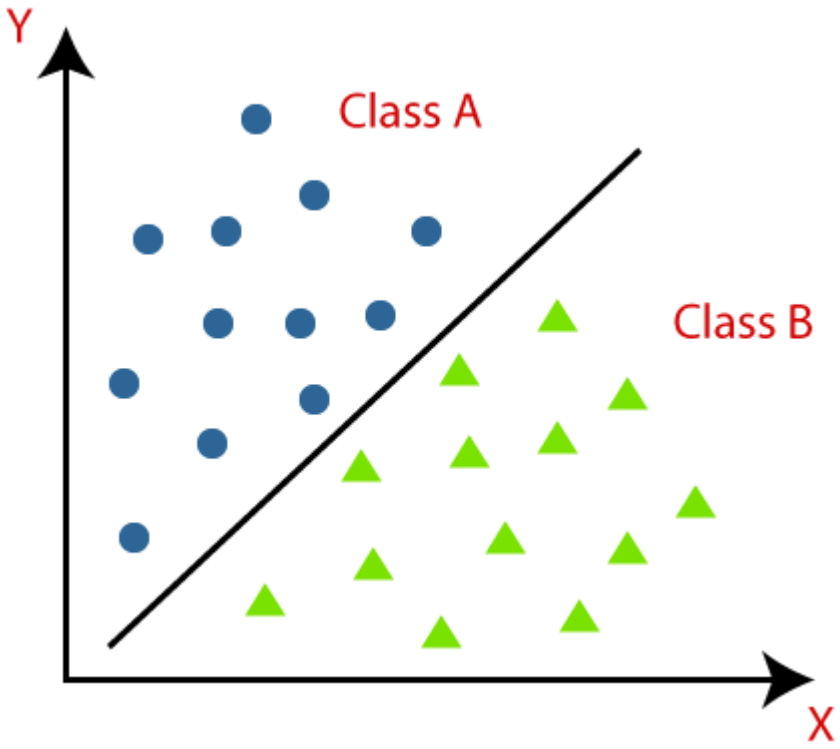


Classification

- A method of machine learning wherein the variable we wish to predict is of a categorical nature.
- The label of a given entry is determined based on known labelled cases.

Supervised or unsupervised?

Supervised!

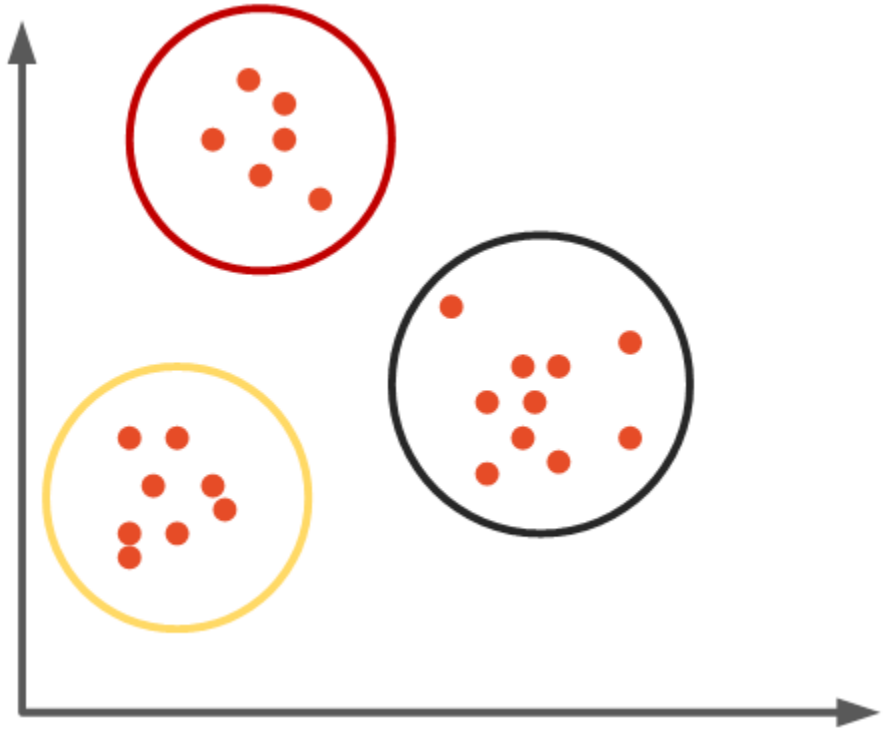


Clustering

- A method of machine learning in which we try to detect clusters of observations in our dataset.
- A given entry is assigned to a group base on the entire dataset.

Supervised or unsupervised?

Unsupervised!



Frequently used libraries

- Pandas - This library is used for structured data operations, like import CSV files, create dataframes, and data preparation
- Numpy - This is a mathematical library. Has a powerful N-dimensional array object, linear algebra, Fourier transform, etc.
- Matplotlib - This library is used for visualization of data.
- Sklearn - This library is used for machine learning tasks. It includes utilities for model training, testing, and validation.









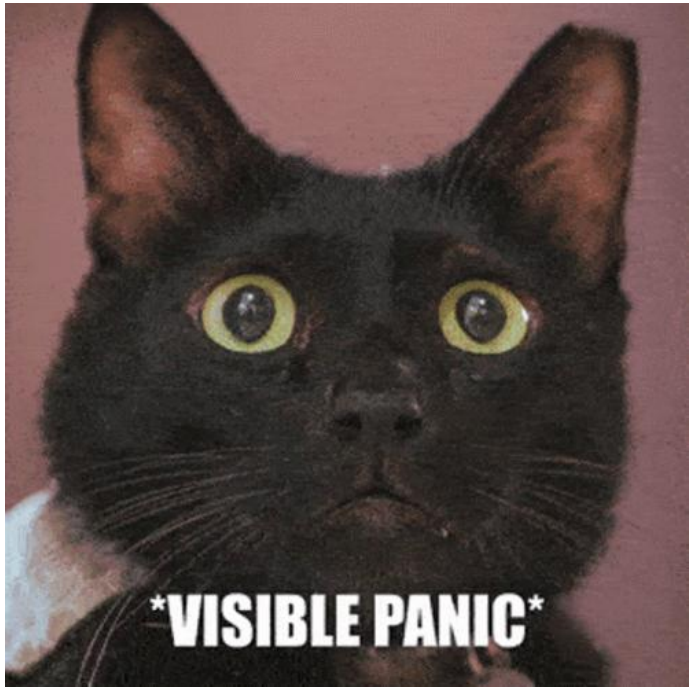
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Let's jump into action

Excercises

Well, almost

- You need to walk before you can learn to run.
- First we need to get everything set up!
- Install Python + JupyterLab



Install python + JupyerLab

- Install Python (https://www.python.org/downloads/release/python-3143/)
  - Add to path if needed (https://phoenixnap.com/kb/add-python-to-path)
  - Try it out! “py --version”
- Install Pip
  - Should be installed by default, if not use https://pip.pypa.io/en/stable/installation/
  - If it doesn’t work, add to path (https://builtin.com/software-engineering-perspectives/pip-command-not-found)
  - Try it out! “pip --version”
- Install JupyterLab (https://jupyter.org/install)
  - Try it out! “jupyter lab” If this doesn’t work, try “jupyter –m lab”

Open a notebook

- Go to canvas and download the notebook “DSF 01 – Intro Notebook”
- Also download the csv file called “DSF 01 – Intro Data”
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run “jupyter lab”
  - Or launch it via anaconda
- Go through the notebook and make the excercises